Retirement and commitments with present-biased preferences

Mario Bernasconi*

Abstract

Present-biased preferences induce dynamically inconsistent decisions, implying a motive for people to constrain their future choices. We present a simple model of savings and retirement to show that sophisticated people, who are aware of their bias, use illiquid assets to constrain future consumption and to prevent retiring earlier than planned. Empirical evidence using survey data from Germany confirms the model predictions. We find that naïve present-biased people retire on average 1.7 years earlier than time-consistent, while fully sophisticated people are most likely to hold illiquid assets and retire 1.7 years later than naïve, thereby completely overcoming their bias.

1 Introduction

Income adequacy in old age is crucial to ensure retirees' well-being, and it largely depends on past saving and retirement decisions. These two decisions are interconnected and share an inter-temporal dimension, meaning that time preferences play an important role. Present-biased preferences, which give stronger relative weight to the near future as it gets closer (O'Donoghue and Rabin, 1999), lead to dynamically inconsistent saving and retirement decisions, creating an incentive for individuals to limit their future options.

In this paper, we study consumption and retirement decisions of present-biased people when savings can be invested in an illiquid asset. We distinguish between naïve people, who are not aware of their present-bias, and sophisticated ones, who are. While naïve people deviate from the planned consumption path and may retire earlier than planned, we show that sophisticated

*Faculty of Business and Economics, University of Basel, Peter Merian-Weg 6 4052 Basel, Switzerland (e-mail: mario.bernasconi@unibas.ch). I thank the participants of Netspar conferences and seminars at Tilburg University and Universitat Autònoma de Barcelona. In particular, Jaap Abbring, Margherita Borella, Jochem de Bresser, Eric French, Jorgo Goossens, Max Groneck, Fabien Ize, Tunga Kantarcı, Gabriella Massenz, Arthur van Soest, Eduard Suari-Andreu, Jeroen van der Vaart, and Conny Wunsch provided helpful comments on an earlier version of this paper. An earlier version of this paper with different results was part of my PhD thesis (Bernasconi, 2024).

people use the illiquid asset to constrain future consumption and labour supply. It is intuitive that low liquidity incentivizes work and constrains consumption, but there may not exist a liquidity level that leads to both first-best consumption and retirement age. For example, the share of illiquid assets that results in an optimal allocation of consumption over time when retirement is exogenous could lead to suboptimal labor supply decisions when retirement is endogenous. Instead, we show that the optimal liquidity level for the case of exogenous retirement also leads to the first-best retirement age when retirement is endogenous.

We further test the model predictions using population-representative survey data from Germany. We exploit the German Socio-Economic Panel's Innovation Sample and categorize people as time-consistent, naïve, partially or fully sophisticated present-biased following the method of Cobb-Clark et al. (2024). In practice, we compare information on their ideal, predicted, and realized body weight to infer their type. First, in line with the model predictions, we find that fully sophisticated people are around 10 to 20 percentage points (around 40 to 90%) more likely to hold illiquid assets, such as private pensions, compared to both naïve and time-consistent people. Second, naïve people retire on average 1.7 years earlier than time-consistent, while fully sophisticated people retire 1.7 years later than naïve, thereby fully overcoming their bias. Third, we find that, conditional on time preferences and sophistication, illiquid assets are positively correlated with later retirement, in line with the idea that they can be used to constrain retirement choices. Additionally, we find that differences in retirement timing across types are smaller among people with illiquid assets, but remain large among those without. However, throughout our analysis, we find that partially sophisticated people behave similarly to naïve individuals, suggesting that for them the perceived benefit of committing is lower than its cost.

Our work contributes to the literature on savings and retirement with present-biased preferences (Phelps and Pollak, 1968; Laibson, 1997; Diamond and Kőszegi, 2003; Schreiber and Weber, 2016; Merkle et al., 2024; Groneck et al., 2024). In particular, Laibson (1997) studies saving decisions of present-biased agents and their demand for illiquid assets while abstracting from labour supply choices. On the other hand, Diamond and Kőszegi (2003) study retirement decisions of present-biased agents when commitments are not available. We combine the insights from these two models in a unified theoretical framework to show that illiquid assets can be used to constrain multiple interconnected decisions, such as consumption and retirement. On the empirical side, we present the first evidence of differences in retirement behaviour between naïve and sophisticated people, and on the mechanism behind these differences. Merkle et al. (2024)

show that present-biased people retire on average 1.75 years earlier than time-consistent ones in Germany. We go beyond their results by distinguishing between naïve, partially, and fully sophisticated present-biased people. Using a different sample and classification method, we find that only naïve and partially sophisticated retire 1.7 years earlier than time-consistent people, while only fully sophisticated are more likely to use commitments compared to time-consistent.

Importantly, our model allows us to study the theoretical *effects* (under some assumptions) of time preferences and sophistication on outcomes, while our empirical analysis presents *correlational* evidence that is consistent with the model predictions. Because it's difficult to exploit exogenous variation in preferences, the literature on time-inconsistency and commitments has focused on correlational evidence, and it has found mixed evidence.[1] For example, Augenblick et al. (2015) and Kaur et al. (2015) find a positive correlation between the demand for commitments and time-inconsistency, while John (2020) and Sadoff et al. (2020) find a negative correlation. Instead, we find a positive correlation between the use of commitments and full sophistication about one's own time-inconsistency. Furthermore, we provide the first evidence that fully sophisticated people are able to overcome their time-inconsistency in the retirement context, and that this result likely stems from the use of commitments.

The remainder of the paper is organized as follows. In section 2, we first solve the model for the cases of exogenous retirement, and then endogenize the retirement decision. In section 3, we introduce an illiquid asset. Section 4 presents the empirical evidence. Section 5 concludes.

2 Model

Our model setup follows Diamond and Kőszegi (2003), but we later add illiquid assets in section 3. Time preferences are modelled with a quasi-hyperbolic discounting function, such that the agent maximizes $u_1 + \sum_{t=2}^{T} \beta\delta^t u_t$. We assume $\beta \in (0,1)$ and $\delta = 1$.[2] We further assume a logarithmic utility over consumption and $T = 3$. In period 1, the agent works and earns $w_1 > 0$, and decides consumption c_1 and savings $s_1 = w_1 - c_1$.[3] In period 2, he decides to work or to retire, and how much to consume (c_2) and save. If he works, he earns $w_2 > 0$ and suffers an additive disutility $e = 1$.[4] In period 3, he cannot work and consumes what's left (c_3). There is

[1]Instead, Carrera et al. (2022) and Westphal (2024) exploit experimental variation in sophistication about time-inconsistent preferences.

[2]As in Laibson (2015), John (2020) and Fahn and Seibel (2022) to simplify the exposition.

[3]w_1 can be interpreted as the sum of initial wealth and period 1 wage.

[4]Results do not qualitatively differ for $e > 0$, but, for example, for large enough values of e retirement is never time-inconsistent. We therefore focus on a simple case where retirement can be time-inconsistent.

no uncertainty and no return on savings.

In section 2.1, we study the consumption decision. In section 2.2, we endogenize the retirement decision. We first review the findings of Diamond and Kőszegi (2003), who showed that for certain levels of savings (s_1) retirement can be time-inconsistent. Differently from their work, we further show that such levels of savings can endogenously originate within the model.[5]

2.1 Consumption decision

The agent's choices can be modelled as an equilibrium in a sequential game played by different selves (self 1 in period 1, self 2 in period 2, etc.). To fix ideas, we first ignore the retirement decision: The agent only works in period 1, earns w_1, and the only choice he makes concerns consumption/savings. For the rest of the paper, we adopt the following notation: We use subscripts to indicate periods, e.g. c_2 is consumption in period 2, and we use asterisks to indicate the self who is deciding, e.g. c_2^{**} is the optimal consumption according to self 2's preferences, while c_2^* is the optimal consumption from the perspective of self 1.

2.1.1 Naïve agent

A naïve self 1 solves the following problem, where period 2 is weighted equally to period 3:

$$\max_{c_1,c_2,c_3} u(c_1) + \beta u(c_2) + \beta u(c_3) \text{ s.t. } c_1 + c_2 + c_3 = w_1$$

which yields $c_1^* = \frac{1}{1+2\beta} w_1$, $c_2^* = \frac{1}{2} s_1^* = \lambda_1 s_1^*$, and $c_3^* = \frac{1}{2} s_1^* = (1 - \lambda_1) s_1^*$, with $s_1^* = \frac{2\beta}{1+2\beta} w_1$ and $\lambda_1 = \frac{1}{2}$ is the share of savings that self 1 would like self 2 to consume. However, self 2 solves the following problem, where the relative weight of period 2 to period 3 ($1/\beta$) is higher compared to the problem above:

$$\max_{c_2,c_3} u(c_2) + \beta u(c_3) \text{ s.t. } c_2 + c_3 = s_1^*$$

which yields $c_2^{**} = \frac{1}{1+\beta} s_1^* = \lambda_2 s_1^*$ and $c_3^{**} = \frac{\beta}{1+\beta} s_1^* = (1 - \lambda_2) s_1^*$, where $\lambda_2 = \frac{1}{1+\beta}$ is the optimal share that self 2 consumes. Since $\beta \in (0, 1)$, $c_2^* < c_2^{**}$ and $c_3^* > c_3^{**}$, and self 2 consumes more than originally planned by self 1.

[5] Our analysis also relates to Holmes (2010), who studies the specific case when $w_1 = w_2$ and shows that retirement cannot be time-inconsistent.

2.1.2 Sophisticated agent

A sophisticated self 1 knows that self 2 consumes c_2^{**} as defined above, regardless of his initial planning. His choice is then the solution of the problem in (1). As shown by Phelps and Pollak (1968), it yields the same optimal saving of naïve self 1, meaning that the realized consumption path of naïve and sophisticated agents is the same.

$$\max_{s_1} u(w_1 - s_1) + \beta u\left(\frac{1}{1+\beta}s_1\right) + \beta u\left(\frac{\beta}{1+\beta}s_1\right) \tag{1}$$

This property of the logarithmic utility simplifies the analysis when retirement is endogenous, and it highlights that the optimal consumption/saving levels of self 1 for period 1 do not depend on the consumption allocation between periods 2 and 3. Consider any two different allocations characterized by λ_i and λ_j. From the perspective of self 1, moving from allocation j to i shifts lifetime utility upwards or downwards (ΔU does not depend on s_1):

$$\begin{aligned} \Delta U &\equiv \beta \ln(\lambda_i s_1) - \beta \ln(\lambda_j s_1) + \beta \ln((1-\lambda_i)s_1) - \beta \ln((1-\lambda_j)s_1) \\ &= \beta \ln\left[\frac{\lambda_i(1-\lambda_i)}{\lambda_j(1-\lambda_j)}\right] \end{aligned} \tag{2}$$

With $\lambda_i = \lambda_1$ and $\lambda_j = \lambda_2$, $\Delta U > 0$ by construction.

2.2 Retirement decision

We now study the retirement decision given the conditional consumption policies outlined above. In particular, self 2 can now work and earn $w_2 > 0$ while suffering a disutility cost $e = 1$.

2.2.1 Naïve agent

Self 2 decides the consumption allocation and consumes a fraction of wealth equal to $\lambda_2 = \frac{1}{1+\beta}$ in period 2. He works if[6]

$$u(\lambda_2(s_1 + w_2)) - u(\lambda_2 s_1) + \beta[u((1-\lambda_2)(s_1+w_2)) - u((1-\lambda_2)s_1)] \geq e \tag{3}$$

[6] We assume that the agent works when indifferent.

Since utility is concave, condition (3) holds for $s_1 \leq \bar{k}_2$, that is the value for which self 2 is indifferent between working or not:

$$\bar{k}_2 \equiv \frac{w_2}{\exp(\frac{1}{1+\beta}) - 1} \tag{4}$$

Consider now self 1's point of view. Since self 1 is naïve, he thinks that self 2 will stick to the allocation that is optimal for self 1 (λ_1) and that self 2 will decide based on the following comparison:

$$\beta[u(\lambda_1(s_1 + w_2)) - u(\lambda_1 s_1)] + \beta[u((1 - \lambda_1)(s_1 + w_2)) - u((1 - \lambda_1)s_1)] \geq \beta e$$

or

$$u(\lambda_1(s_1 + w_2)) - u(\lambda_1 s_1) + [u((1 - \lambda_1)(s_1 + w_2)) - u((1 - \lambda_1)s_1)] \geq e \tag{5}$$

The difference between (3) and (5) is just the factor $\beta \in (0, 1)$, which generates time-inconsistent preferences, while the different λs are innocuous. The left-hand side of (5) is greater than that of (3), and naïve self 1 thinks that the threshold that self 2 will use to decide is

$$\bar{k}_1 \equiv \frac{w_2}{\exp(\frac{1}{2}) - 1}, \tag{6}$$

with $\bar{k}_1 > \bar{k}_2$. For $\bar{k}_2 < s_1 \leq \bar{k}_1$, self 2 retires, but naïve self 1 thinks that self 2 will work. Expanding on Diamond and Kőszegi (2003), we note that the saving level is not exogenous, but it comes from the choice of self 1. Naïve self 1 solves the following problem:

$$\begin{aligned}
s_1^* &= \arg\max_{s_1} U(s_1) \\
&= \arg\max_{s_1} u(w_1 - s_1) \\
&\quad + \beta[u(\lambda_1 s_1) + u((1 - \lambda_1)s_1)] \times \mathbb{1}\{s_1 > \bar{k}_1\} \\
&\quad + \beta[u(\lambda_1(s_1 + w_2)) + u((1 - \lambda_1)(s_1 + w_2)) - e] \times \mathbb{1}\{s_1 \leq \bar{k}_1\} \\
&= \max_{s_1} \; U^w(s_1) \times \mathbb{1}\{s_1 > \bar{k}_1\} + U^r(s_1) \times \mathbb{1}\{s_1 \leq \bar{k}_1\}
\end{aligned} \tag{7}$$

where $U(s_1)$ is the lifetime utility of naïve self 1, $U^w(s_1)$ is the lifetime utility conditional on working in period 2, and $U^r(s_1)$ conditional on retiring in period 2. The savings level that

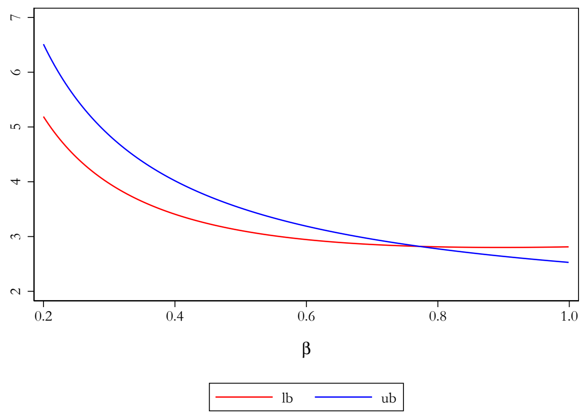

Figure 1: Bounds on w_1/w_2 for time-inconsistent retirement as a function of β, see (10).

maximizes the conditional utilities $U^w(s_1)$ and $U^r(s_1)$ are given by s_1^w and s_1^r (and do not depend on λ_1, see (2)):

$$s_1^w = \frac{2\beta w_1 - w_2}{1 + 2\beta}, \tag{8}$$

$$s_1^r = \frac{2\beta w_1}{1 + 2\beta}. \tag{9}$$

In order to have time-inconsistency we need that (i) self 1 wants to work and (ii) self 2 prefers to retire given self 1 optimal savings decision s_1^w. That is equivalent to (i) $U^w(s_1^w) \geq U^r(s_1^r)$ (which implies $s_1^* = s_1^w$ and $s_1^w \leq \bar{k}_1$) and (ii) $\bar{k}_2 < s_1^w$. These two conditions boil down to a condition on the realized state $\{w_1, w_2\}$. In particular, the ratio w_1/w_2 needs to be low enough such that self 1 wants to work but high enough such that self 2 retires, and the bounds on this ratio, lb and ub, depend on time preference β (see (10)). Figure 1 plots these bounds as function of β, showing the possibility of time-inconsistent retirement: For certain values of β it holds that $ub > lb > 0$, such that (10) is satisfied for some positive w_1, w_2.

$$lb \equiv \frac{1}{2\beta}\left[\frac{1+2\beta}{\exp(\frac{1}{1+\beta}) - 1} + 1\right] < \frac{w_1}{w_2} \leq \frac{1}{\exp(\frac{\beta}{1+2\beta}) - 1} \equiv ub \tag{10}$$

We can therefore distinguish three cases. Case 1: when $w_1/w_2 < ub$ and $w_1/w_2 < lb$, self 1 and self 2 agree to work in period 2. Case 2: when $w_1/w_2 > ub$, self 1 and self 2 agree to retire in period 2. Case 3: when $lb < w_1/w_2 \leq ub$, self 1 would like to work in period 2, but self 2 retires given self 1's first-best savings.

2.2.2 Sophisticated agent

A sophisticated self 1 knows that self 2 chooses the consumption allocation for periods 2 and 3 and that self 2 works if $s_1 \leq \bar{k}_2$. This means that self 1 can influence the time of retirement by choosing the level of savings that self 2 will inherit. We formalize sophisticated self 1's problem below:

$$
\begin{aligned}
\max_{s_1} U(s_1) = \max_{s_1} \; & u(w_1 - s_1) \\
& + \beta[u(\lambda_2 s_1) + u((1-\lambda_2)s_1)] \times \mathbb{1}\{s_1 > \bar{k}_2\} \\
& + \beta[u(\lambda_2(s_1 + w_2)) + u((1-\lambda_2)(s_1 + w_2)) - e] \times \mathbb{1}\{s_1 \leq \bar{k}_2\} \\
= \max_{s_1} \; & U^w(s_1) \times \mathbb{1}\{s_1 > \bar{k}_2\} + U^r(s_1) \times \mathbb{1}\{s_1 \leq \bar{k}_2\}
\end{aligned} \tag{11}
$$

(with λ_2 and $\bar{k}_2$ instead of λ_1 and $\bar{k}_1$ compared to (7)). In particular, because of (2), the savings that maximise $U^w(s_1)$ and $U^r(s_1)$ are the same for naïve and sophisticated (see (8) and (9)).

Because of the concavity of utility, the solution of (11) is one of the following: s_1^r, s_1^w or $\bar{k}_2$. Assume now that $U^w(s_1^w) > U^r(s_1^r)$, meaning that self 1 first-best is to save s_1^w and work in $t = 2$. If $\bar{k}_2 < s_1^w$, self 1 instead saves $\bar{k}_2$ or s_1^r. As shown in Diamond and Kőszegi (2003), if $U^w(\bar{k}_2) \geq U^r(s_1^r)$, self 1 under-saves in order to induce self 2 to work since he cannot reach $U^w(s_1^w)$. On the other hand, if $U^w(\bar{k}_2) < U^r(s_1^r)$, self 1 over-saves to compensate for the early retirement decision of self 2. However, differently from Diamond and Kőszegi (2003), we show next that a sophisticated self 1 can use an illiquid asset to reach his first-best retirement age and consumption allocation, with no need to over or under-save.

3 Commitment device

3.1 Consumption decision

In this section, we only consider a sophisticated agent, because a naïve agent would not use commitments. Again, we start with the case of exogenous retirement. Assume now that an illiquid asset is available. In period 1, self 1 decides to invest a share α of his savings in the liquid asset and $(1-\alpha)$ in the illiquid one. Both assets do not yield any return.[7] In period 2, self 2 has at his disposal only the liquid savings αs_1.[8] In period 3, self 3 consumes what is left,

[7]The qualitative results do not depend on the identical returns assumption.

[8]We assume, as in Laibson (1997), that if the agent sells the illiquid asset in period 2 he would get paid in period 3. If he applies for a loan at time 2, the associated cash will not be available for consumption until time

that is $\alpha s_1 - c_2 + (1-\alpha)s_1$.

The problem of self 2 is now different because he cannot consume his preferred quantity if this is larger than its liquidity. This is formalized by a liquidity constraint (LC) on top of the budget constraint (BC):

$$\max_{c_2,c_3} u(c_2) + \beta u(c_3)$$

$$\text{s.t. } c_2 \leq \alpha s_1 \qquad \text{(LC)}$$

$$c_3 = \alpha s_1 - c_2 + (1-\alpha)s_1 \qquad \text{(BC)}$$

which yields

$$c_2^{**} = \begin{cases} \lambda_2 s_1 & \text{if } \lambda_2 s_1 \leq \alpha s_1 \\ \alpha s_1 & \text{otherwise} \end{cases}, \tag{12}$$

$$c_3^{**} = \begin{cases} (1-\lambda_2)s_1 & \text{if } \lambda_2 s_1 \leq \alpha s_1 \\ (1-\alpha)s_1 & \text{otherwise} \end{cases}.$$

This shows how self 1 can constrain the choice of self 2, $\{c_2^{**}, c_3^{**}\}$, through α. We have shown above that self 1 would like self 2 to consume $c_2^* = \lambda_1 s_1 = \frac{1}{2}s_1$, and his optimal savings s_1^r are given by (9). With log utility, the choice of c_1/s_1 is not affected by the allocation between periods 2 and 3. Therefore, the illiquid asset shifts upward the lifetime utility of the agent, because the consumption allocation between periods 2 and 3 is improved and the optimal c_1/s_1 does not change. Self 1 sets $\alpha^* = \lambda_1 = \frac{1}{2} < \frac{1}{1+\beta} = \lambda_2$ and $s_1^* = s_1^r$. In this way, self 2 is forced to consume all the liquid savings in period 2 and all the illiquid ones in period 3, where these consumption levels correspond to the first-best of self 1.

3.2 Retirement decision

We now introduce the retirement decision in period 2. Self 2's consumption plan if he retires is given by (12). We refer to this plan as (c_2^{r**}, c_3^{r**}). Self 2's problem conditional on working,

period 3.

instead, is given by:

$$\max_{c_2,c_3} u(c_2) - e + \beta u(c_3)$$

$$\text{s.t. } c_2 \leq \alpha s_1 + w_2 \qquad \text{(LC)}$$

$$c_3 = \alpha s_1 + w_2 - c_2 + (1-\alpha)s_1 \qquad \text{(BC)}$$

which yields

$$c_2^{w**} = \begin{cases} \lambda_2(s_1 + w_2) & \text{if } \lambda_2(s_1+w_2) \leq \alpha s_1 + w_2 \\ \alpha s_1 + w_2 & \text{otherwise} \end{cases}, \tag{13}$$

$$c_3^{w**} = \begin{cases} (1-\lambda_2)(s_1 + w_2) & \text{if } \lambda_2(s_1+w_2) \leq \alpha s_1 + w_2 \\ (1-\alpha)s_1 & \text{otherwise} \end{cases}.$$

Given s_1 and α, self 2 computes $\{c_2^{r**}(s_1,\alpha), c_3^{r**}(s_1,\alpha)\}$ and $\{c_2^{w**}(s_1,\alpha), c_3^{w**}(s_1,\alpha)\}$. Self 2 decides to work or not by comparing the difference in utilities from the two consumption plans and the disutility e. The choice of self 1 regarding s_1 and α influences both future consumption and retirement. We now analyse the three cases in which self 1 and self 2 agree or disagree on the retirement decision.

Case 1: $w_1/w_2 < ub$ and $w_1/w_2 < lb$ (Self 1 and self 2 agree to work in period 2 given self 1's first-best saving s_1^w and no commitment.)

Self 1 sticks to s_1^w and sets $\alpha = \alpha^w$ to reallocate consumption from period 2 to 3:

$$\lambda_1(s_1^w + w_2) = \frac{1}{2}(s_1^w + w_2) = \alpha^w s_1^w + w_2 \implies \alpha^w = \frac{\frac{1}{2}(s_1^w + w_2) - w_2}{s_1^w} \tag{14}$$

Appendix 6.1 shows that self 2 would still prefer to work given α^w and s_1^w. This result implies that self 1 can reach his first-best consumption allocation and retirement age.

Case 2: $w_1/w_2 > ub$ (Self 1 and self 2 agree to retire in period 2 given self 1's first-best saving s_1^r and no commitment.)

Self 1 sticks to s_1^r and sets $\alpha = \alpha^r$ to reallocate consumption from period 2 to 3:

$$\lambda_1 s_1^r = \frac{1}{2}s_1^r = \alpha^r s_1^r \implies \alpha^r = \frac{1}{2} \tag{15}$$

Appendix 6.2 shows that self 2 would still prefer to retire given α^r and s_1^r. Also for this case, the result implies that self 1 can reach his first-best consumption allocation and retirement age.

Case 3: $w_1/w_2 \in (lb, ub]$ (Self 1 and self 2 disagree on the retirement decision given self 1's first-best saving s_1^w and no commitment.)

In order to force self 2 to work, self 1 has to set a low α. Similarly, in order to improve the consumption allocation, he has to set a low α. Self 1 sets $s_1 = s_1^w$, which induces retirement if no commitment is provided, and the same α^w as in case 1. Appendix 6.3 shows that self 2 prefers to work given savings s_1^w and a share of liquid savings α^w if $w_1/w_2 \in (lb, ub]$. Also for this case, the result implies that self 1 can reach his first-best consumption allocation and retirement age.

3.3 Discussion

We presented a stylized model of consumption and retirement with some simplifying assumptions. Before testing the model against empirical data — where these assumptions may not hold — it is useful to reflect on their implications for the main results.

Laibson (2015) emphasizes that as future uncertainty increases, the value of flexibility rises at the expense of the value of commitment. In the context of our model, uncertainty related to future income (encompassing unemployment risk), uncertain expenses (such as medical ones), or uncertain disutility form working (e.g. due to uncertain health conditions) are perhaps most interesting. Also in our model, uncertainty lowers the marginal utility from using illiquid assets, because the commitment cannot be targeted to different states of the world. Still, in the model extension in Appendix 6.4, we show that targeting the share of liquid and illiquid assets to the worst-case scenario is always better than holding no illiquid assets at all even with uncertainty. Lower benefits from the commitment, potentially reduced also by the presence of direct costs associated with the take up of the commitment (e.g. the hassle of setting up a private pension account, management fees, etc.), imply that sophisticated individuals are less likely to hold illiquid assets than they would in a setting without uncertainty. Nevertheless, sophisticated agents would still be more likely to use commitments than naïve ones, which is the prediction that we will test.

The assumption of deterministic and identical returns of liquid and illiquid assets might also be restrictive. If returns are uncertain, naïve and time-consistent individuals might still choose to hold some illiquid assets for risk mitigation purposes. The same applies if illiquid assets yield systematically higher returns (e.g. liquidity premium). However, sophisticated individuals would continue to perceive an additional benefit from holding illiquid assets due to their value as commitment devices. As a result, they would still be more likely to allocate a share of their savings to illiquid assets compared to naïve and time-consistent people.

Finally, we have so far assumed that people are either fully sophisticated or naïve about their present-bias. Partially sophisticated people fall between these two extreme cases. Following O'Donoghue and Rabin (2001), partially sophisticated individuals are aware of their present-bias, but they underestimate it. In practice, a partially sophisticated self 1 thinks that self 2 discounts future utility by a factor $\hat{\beta}$, with $\beta < \hat{\beta} < 1$. Therefore, partially sophisticated self 1 thinks that self 2 consumes more than self 1's optimal share, but not as much as self 2 actually consumes. Partially sophisticated self 1 also thinks that self 2 decides to retire based on $\hat{\beta}$, meaning that he might end up retiring earlier than planned.

A partially sophisticated agent values commitments, but less than a fully sophisticated one. If the commitment is free, partially and fully sophisticated people behave in the same way in our model.[9] If the commitment comes at a positive price, there is a range of values of the price for which fully sophisticated people commit but partially sophisticated don't. Therefore, we further distinguish between these two types in the empirical analysis.

4 Empirical evidence

4.1 Hypotheses

In our stylized model, fully and partially sophisticated agents use illiquid assets to commit their future selves, while naïve agents (and time-consistent) don't. In the data, we do not expect sophisticated people to *always* hold illiquid assets, because they might use alternative commitments, or because uncertainty and associated costs make their use less appealing as explained above. Similarly, we do not expect naïve and time-consistent people to *never* hold

[9] Heidhues and Kőszegi (2009); John (2020); Bai et al. (2021); Carrera et al. (2022) argue that partially sophisticated individuals may be sophisticated enough to use commitments, but they might also adopt weak commitment devices that are ineffective, ultimately harming themselves if the commitment is costly. Sophisticated agents would then prefer not to commit if commitments are weak. In the model we presented, the commitment cannot be weak because self 2 can never access the illiquid savings, as in Laibson (1997). The commitment could be weak if self 2 can pay a (small enough) penalty to liquidate the illiquid asset. In line with our theoretical framework, the empirical analysis focuses on commitments that involve substantial penalties, which are unlikely to be weak, or that are entirely illiquid. This implies that the commitments are unlikely to fail, and a fully sophisticated agent would not turn them down.

illiquid assets, because they might be useful for reasons other than commitment. We also expect partially sophisticated people to commit less often than fully sophisticated individuals, because commitments come at a positive cost (at least in terms of transaction or hassle costs).

Nevertheless, we expect the model's predictions regarding differences between types to hold on average. This yields a testable ordering in the likelihood of holding illiquid assets: fully sophisticated individuals should be the most likely to hold such assets, followed by partially sophisticated individuals, and lastly by naïve and time-consistent individuals. Consider the regression equation in (16), where $Illiquid$ is a dummy for holding illiquid assets, TI is a dummy for being time-inconsistent, PS (FS) is a dummy for being partially (fully) sophisticated, and X is a vector of controls (which we detail later). The coefficient θ_1 measures differences between naïve and time-consistent people, θ_2 (θ_3) measures differences between partially (fully) sophisticated and naïve people.

$$Illiquid_i = \theta_0 + \theta_1 TI_i + \theta_2 PS_i + \theta_3 FS_i + \theta_4 X_i + \varepsilon_i \tag{16}$$

We state our first hypothesis in terms of coefficients from (16):

Hypothesis 1. $\theta_1 = 0$ *and* $\theta_3 > \theta_2 > 0$ *in (16).*

The first hypothesis is also consistent with a model of exogenous retirement, where illiquid assets are used to constrain consumption. However, our model also predicts a testable – and different – ordering of types with respect to their retirement age. Consider the regression equation in (17), where we regress retirement age on the same covariates as above.

$$RetAge_i = \theta_0 + \theta_1 TI_i + \theta_2 PS_i + \theta_3 FS_i + \theta_4 X_i + \varepsilon_i \tag{17}$$

Our model shows that, conditional on the saving level, naïve present-biased people retire sometimes earlier than time-consistent, but never later. We thus expect, on average, the retirement age of time-consistent people to be higher compared to naïve individuals, i.e. $\theta_1 < 0$ in (17). The model also predicts that sophisticated people use the illiquid asset to avoid retiring earlier than planned, meaning that we expect sophisticated people to retire later than naïve ones, and more so if they are fully rather than partially sophisticated ($\theta_3 > \theta_2 > 0$ in (17)). Putting things together, we state the second hypothesis as follows:

Hypothesis 2. $\theta_1 < 0$ *and* $\theta_3 > \theta_2 > 0$ *in (17).*

Moreover, if fully sophisticated people manage to completely overcome their bias, as the model suggests, we should expect the sum of the coefficients θ_1 and θ_3 to be close to zero.

The first two hypotheses are also consistent with a model in which people use illiquid assets to constrain consumption and a second commitment device to constrain retirement. However, a distinctive aspect of our model is that illiquid assets act as commitments to constrain retirement decisions. We test this implication in two ways. First, we investigate whether, conditional on time-preferences and sophistication, illiquid assets are positively correlated with later retirement by estimating equation (18). Second, we re-estimate equation (17) by splitting the sample based on illiquid asset holdings, as specified in equations (19) and (20). The underlying idea is that if illiquid assets cause present-biased people to retire later than they would otherwise, and if sophisticated people exploit this mechanism when necessary, the difference in retirement age between types should be smaller for the sub-sample of people with illiquid assets (19) compared to those without (20).

$$RetAge_i = \theta_0 + \theta_1 TI_i + \theta_2 PS_i + \theta_3 FS_i + \theta_4 X_i + \theta_5 Illiquid_i + \varepsilon_i \tag{18}$$

$$RetAge_i = \check{\theta}_0 + \check{\theta}_1 TI_i + \check{\theta}_2 PS_i + \check{\theta}_3 FS_i + \check{\theta}_4 X_i + v_i \quad \text{if} \quad Illiquid_i = 1 \tag{19}$$

$$RetAge_i = \tilde{\theta}_0 + \tilde{\theta}_1 TI_i + \tilde{\theta}_2 PS_i + \tilde{\theta}_3 FS_i + \tilde{\theta}_4 X_i + u_i \quad \text{if} \quad Illiquid_i = 0 \tag{20}$$

We thus state a third hypothesis as follows:

Hypothesis 3. $\theta_5 > 0$ *in (18).* $0 > \check{\theta}_1 > \tilde{\theta}_1$, $\tilde{\theta}_2 > \check{\theta}_2 > 0$ *and* $\tilde{\theta}_3 > \check{\theta}_3 > 0$ *in (19) and (20).*

4.2 Data and classification of types

In order to test the implications of our model, we need information at the individual level on time preferences and on choices regarding the take-up of illiquid assets and retirement. Because preferences and sophistication levels are not observed, we have to infer them. To this end, we exploit the German Socio-Economic Panel's Innovation Sample (SOEP-IS, Richter and Schupp 2015) and the classification method from Cobb-Clark et al. (2024)(CC from now on).[10] The main advantage of this method is that it not only allows us to categorize respondents as time-consistent or inconsistent, but further as naïve, partially, or fully sophisticated when time-inconsistent. Moreover, as explained below, the method does not rely on information regarding

[10] The data we use differs slightly from that in CC, who had early access to a preliminary version while we use the publicly released one.

commitments take up, investment decisions, or work behavior, and is thus not mechanically correlated with the outcomes of interest.

The SOEP-IS is a population representative annual household panel study comprising around 5,500 people. It is designed for methodical and thematic innovative research and collects somewhat different information compared to the SOEP-Core.[11] As a result, the same questions are not asked in every wave. Moreover, not all SOEP-IS participants are administrated the same modules. The authors of CC designed new questions for the 2017 and 2018 waves that would allow them to elicit time preferences and sophistication. In 2017, a subset of 1,780 respondents were asked to report their current, ideal and predicted future weight one year later.[12] In 2018, they were asked to report their current weight. Exploiting these questions, CC classify individuals who wish to maintain or lose weight as described in Table 1.

The classification method is consistent with our theoretical modeling of time preferences and sophistication. Time-consistent self 1 ($\beta = \hat{\beta} = 1$) knows that he will stick to his first-best plan (predicted weight = ideal = actual). Present-biased naïve self 1 ($\beta < \hat{\beta} = 1$) thinks that he will stick to his plan (predicted = ideal) but will later deviate from it (actual > predicted). Present-biased partially sophisticated self 1 ($\beta < \hat{\beta} < 1$) knows that he will not reach his first-best (predicted > ideal) but under-estimate his bias (actual > predicted). Present-biased fully sophisticated self 1 ($\beta = \hat{\beta} < 1$) knows that he will not reach his first-best and is fully aware of his bias (actual = predicted > ideal). The classification of fully sophisticated individuals assumes they are unable to overcome their bias. However, if some of these individuals can in fact overcome their bias, they would be incorrectly classified as time-consistent. More broadly, our classification may be subject to errors. Nonetheless, as long as it is moderately positively correlated with the true types, miss-classification results in attenuation bias, which makes it more difficult to detect evidence supporting our hypotheses (see Mirenda et al., 2022 for a similar argument).

We use the 2021 version of the SOEP-IS data and closely follow the classification and sample restrictions of CC, which we report here for the sake of completeness.[13] We start with a sample of 1,647 individuals for which information regarding actual, ideal and predicted weight are not

[11] Originally part of the SOEP-Core, the SOEP-IS has been managed as a separate study since 2011. The earliest available variables in the SOEP-IS date back to 1998.

[12] Information regarding the ideal and predicted weight has not been elicited in other waves. See Table 5 for the exact questions.

[13] Following CC, we classify people as time-consistent or fully sophisticated even when the predicted weight is up to 2 percent lower than then actual weight in 2018, to account for rounding error and the uncertainty about future weight.

Type	Predicted (in 2017 for 2018) vs. Ideal Weight (in 2017 for 2018)	Actual (in 2018) vs. Predicted Weight (in 2017 for 2018)
Time-consistent	Predicted = Ideal	Actual = Predicted
Naïve	Predicted = Ideal	Actual > Predicted
Partially sophisticated	Predicted > Ideal	Actual > Predicted
Fully sophisticated	Predicted > Ideal	Actual = Predicted

Table 1: Classifications rules.

missing. We then focus on people who wish to maintain or lose weight (92% of the sample). We exclude 44 people whose weight changed by more than 10 kg between 2017 and 2018, as this likely reflects health conditions or pregnancies. We exclude 237 individuals whose predicted (38 people) or actual weight in 2018 (199) is below the ideal one. This is because, given that they want to either maintain or lose weight, their behaviour is not in line with utility-maximization under the assumption that losing weight is costly. We finally drop 115 people who, in 2018, weigh less than predicted, because they were overly pessimistic and not covered by our theoretical framework ($\hat{\beta} < \beta$).

The last row of Table 2 reports the distribution of types for the remaining 1,123 people. We find that the majority of people (75%) are time-inconsistent, and that the majority of time-inconsistent people (56%) are at least partially aware of it. These two conclusions are mostly consistent with the few studies that attempt to classify the sophistication degree of people at the individual level (see Table 6).

4.3 Variable definitions

The SOEP-IS data has the unique advantage of allowing us to classify a population representative sample according to their time preferences and sophistication degree. However, as explained above, the additional available variables are limited and not consistent across waves. Therefore, additional information for our sample is scarcer than one might expect.

To test the first hypothesis, we need information on whether individuals hold a financial product that is mostly illiquid and can be used to finance consumption in old age. To capture this, we construct a dummy variable equal to one if the respondent reported having an automated saving plan in the 2016 wave of the survey. Specifically, participants were asked whether they held any of the following: personal pension schemes with state subsidies (Riester- or Rürup-Rente), other personal pension schemes, building savings contracts, cash-value life insurance policies, or capital formation savings plans. Unfortunately, we do not observe which specific product each individual holds. Nevertheless, all of these products are inherently illiquid and designed to prevent over-consumption at young age, thereby financing consumption at older age – even if in the form of housing for the case of building savings contracts.[14] These plans also exhibit features commonly associated with commitment devices. Contributions are often automatically deducted from individuals' income, management fees are charged, and penalties apply if the savings plan is terminated early. In particular, government subsidies must be repaid if a Riester-Rente plan is ended early, and this is also typically the case for capital formation savings plans and building savings contracts, while Rürup-Rente is even less flexible and allows early access to savings only in case of severe disability, thus providing a strong commitment.

Additionally, in 2012, a specific question asked respondents whether they held a Riester- or Rürup-Rente private pension plan. Unfortunately, this variables is available only for 17% of our sample, but can still be used to provide an additional test for the first hypothesis.

To test the second hypothesis, we use information on individuals' retirement timing. Specifically, we define a binary outcome indicating whether a respondent was in the labour force in the last year in which this information was elicited in 2020.[15] Conditional on the year of birth, this allows us to examine whether naïve present-biased individuals retire earlier than other types. We focus on labour force participation rather than employment status to account for potential differences in unemployment risk across types. Moreover, exits from the labour force are more likely to be permanent and thus more indicative of retirement.

Additionally, information regarding the last year of employment is available for some of the individuals in our sample. Complementing this variable with information regarding the labour force participation through the SOEP-IS waves, we impute the retirement age as the difference between the last year of employment and the year of birth. We can thus also use this variable to test the second hypothesis.

With these variables at hand, we estimate equations (16) and (17) while controlling for gender, year of birth, weight in 2017, height, health status, overall monthly savings, self-assessed patience, crystallized intelligence (defined as in Lang et al. 2007), state of residence fixed-

[14] It's also worth noting that all the listed items are likely to be acquired before, and not after, retirement. This is clearly the case for pension schemes (given their scope) and capital formation savings plans (for which the employers pay the contributions in Germany). But it's also likely for building savings contracts (average age of first-time buyers in Germany was around 30, Angelini et al. 2013) and cash-value life insurances (which in Germany typically pay a lump-sum or annuity at retirement if the policyholder survives, making them attractive to acquire before retirement).

[15] While we use the 2021 version of the SOEP-IS data, released in 2023, data collection was suspended for the year 2021.

affects, and month of interview fixed-effects. Table 5 provides more information on the variable definitions and reports the year in which they were assessed.

To test the third hypothesis, we mainly exploit the variables for having an illiquid saving plan in 2016 and for labour force participation in 2020. In particular, we investigate whether illiquid assets are positively correlated with later retirement (equation (18)) and whether the probability of being retired differs by type and by illiquid assets holding (equations (19) and (20)). While in theory we could repeat the latter exercise with retirement age as an outcome, restricting the analysis to retirees who had a saving plan in 2016 leaves us with a small sample (only 67 individuals). We therefore interact the dummies for the different types with the dummy for having a saving plan. Still, the identification of the coefficients for the interaction terms relies on a limited number of observations.

It is worth noting at this point that our classification of the four types is based on data from 2017 and 2018, while the outcome variables used in our tests mainly reflect decisions made in other periods. This raises a potential concern: if time preferences change over time, we might fail to observe the expected correlations. However, evidence from Meier and Sprenger (2015) suggests that time discounting and present bias are stable over time, which mitigates this concern.

Table 2 presents summary statistics by type and for the whole sample for the four outcome variables described above and for the control variables that we include in the regressions (except state of residence and month of interview). The table already suggests that, unconditionally, fully sophisticated individuals are most likely to hold illiquid assets, followed by partially sophisticated, and then by time-consistent and naïve with similar shares. Furthermore, the averages of the dummies 'Saving plan in 2016' and 'Private pension in 2012' are similar for all types, suggesting that Riester- and Rürup-Rente plans are the largest components of the variable 'Saving plan in 2016'. We also see that fully sophisticated people retire approximately at the same age of time-consistent people, while naïve and partially sophisticated people retire earlier.[16] These observations are mostly consistent with our hypotheses, which we now test more formally.

[16]While the average retirement age might seem low by today's standard, the average year of birth for retirees in our sample is 1947, for which early retirement was the norm in Germany (Börsch-Supan and Jürges, 2006).

	Time-consistent	Naive	Part. Soph.	Fully Soph.	Total
Saving plan in 2016 (0/1)	0.24	0.22	0.28	0.37	0.27
	(0.43)	(0.42)	(0.45)	(0.48)	(0.44)
Private pension in 2012 (0/1)	0.20	0.23	0.33	0.40	0.27
	(0.40)	(0.42)	(0.47)	(0.50)	(0.45)
Retirement age	60.06	57.35	56.05	60.25	58.39
	(5.36)	(6.54)	(7.44)	(5.19)	(6.41)
Labour force in 2020 (0/1)	0.55	0.62	0.70	0.68	0.63
	(0.50)	(0.49)	(0.46)	(0.47)	(0.48)
Female (0/1)	0.53	0.50	0.58	0.49	0.53
	(0.50)	(0.50)	(0.49)	(0.50)	(0.50)
Year of birth	1957.29	1963.54	1967.32	1961.73	1962.67
	(16.31)	(17.61)	(16.84)	(17.51)	(17.44)
Monthly savings in 2006	384.82	299.68	307.42	463.73	349.15
	(663.61)	(527.10)	(486.42)	(595.11)	(568.07)
Weight in 2017 (kg)	72.02	80.05	86.65	80.84	79.89
	(13.06)	(16.11)	(17.96)	(14.98)	(16.58)
Height (cm)	170.78	171.23	171.35	171.57	171.20
	(8.86)	(8.92)	(9.44)	(8.99)	(9.05)
Health status in 2017 (1-5)	2.44	2.56	2.74	2.56	2.58
	(0.86)	(0.95)	(0.94)	(0.89)	(0.92)
Patience (standardized)	0.10	0.04	-0.05	-0.16	0.00
	(0.93)	(1.03)	(1.02)	(1.00)	(1.00)
Crystallized intelligence (standardized)	0.19	-0.16	-0.12	0.21	0.00
	(0.99)	(1.04)	(1.03)	(0.82)	(1.00)
Individuals	281	370	293	179	1,123
Share	25%	33%	26%	16%	100%

Table 2: Summary statistics.

Note: Std. deviations in parentheses. The sample size is smaller for the variables 'Private pension in 2012' and 'Retirement age'.

4.4 Results

Table 3 presents the estimates for regressions (16) and (17). Estimates in columns 1 and 2 are consistent with Hypothesis 1: Naïve people are as likely as time-consistent individuals to hold illiquid assets, while fully sophisticated people are significantly more likely (by around 10 or 20 percentage points (p.p.) depending on the outcome definition). These differences are also economically significant given that, across the whole sample, 27% of people holds illiquid assets (see Table 2). The coefficients for partial sophistication are also positive and in line with our hypothesis, i.e. smaller than that for fully sophisticated, but not statistically significant.

Estimates in columns 3 and 4 are consistent with Hypothesis 2. Conditional on being born in the same year, naïve people are 6 p.p. less likely to be in the labour force by 2020 compared to time-consistent, i.e. they retire earlier on average. On the other hand, fully sophisticated individuals are 8 p.p. more likely to be in the labour force than naïve, i.e. they retire later. Consistently, we find that naïve people retire on average 1.70 years earlier than time-consistent, while fully-sophisticated retire approximately 1.66 years later than naïve. The former estimate is very close to the result of Merkle et al. (2024) that time-inconsistent participants retire on average 1.75 years earlier than time-consistent in Germany. Remarkably, the sum of our estimates for θ_1 and θ_3 is very close to zero for both column 3 and 4, suggesting that fully sophisticated people completely overcome their bias and retire at the same age of time-consistent people, as predicted by the model. Still, columns 1 to 4 highlight that, despite similar retirement behaviour, fully sophisticated and time-consistent people differ in their use of commitments, as implied by our model.

Our empirical finding that fully sophisticated people retire at the same age of time-consistent people (while naïve retire early) is potentially in line with two explanations: either sophisticated people do not use illiquid assets and under-save to induce later retirement (as in Diamond and Kőszegi (2003), see Section 2.2.2), or they use illiquid assets as commitment devices and do not deviate from the first-best level of savings. In our theoretical model, the latter option is optimal for sophisticated Self 1. In our empirical setting, illiquid assets are available and they come at a limited cost (managements fees, hassle costs). Therefore, we expect people to use commitments rather than under-saving. While we cannot exclude that *some* sophisticated people under-save, this seems unlikely to be the main explanation given the estimates in columns 1 and 2 of Table 3. Moreover, the unconditional average monthly savings reported in Table 2 do not suggest that fully sophisticated people under-save. In fact, they save most compared to the other types. This holds also when using monthly savings as outcome in our baseline regression model (not shown, $p < 0.01$). The test of Hypothesis 3, which we discuss next, also supports the notion that sophisticated individuals use illiquid assets as a commitment device rather than under-saving.

Concerning the third hypothesis, column 1 in Table 4 presents estimates of equation (18). In particular, we find that illiquid assets are positively correlated with later retirement, conditional on time-preferences and sophistication and the usual controls. The result is consistent with the idea that illiquid assets cause people to retire later. Columns 2 and 3 present estimates for equations (19) and (20), respectively. Column 2 shows that, among people with illiquid assets, the timing of retirement does not differ by type, contrary to what we showed in column 3 of Table 3 for the whole sample. Again, this is in line with the idea that illiquid asset constrain retirement options and align the timing of retirement of time-consistent and inconsistent people. On the other hand, column 3 shows that the differences across types remain significant among people without illiquid assets. For naïve individuals, this shows that if they do not hold illiquid assets, they exit the labour force earlier than time-consistent individuals. For fully sophisticated, however, we find no difference compared to time-consistent (-0.08 + 0.09 close to zero) even when they do not hold illiquid assets. Therefore, either they use commitments that are not captured by our variable, or they do not need commitments to constrain the retirement choice (cases 1 and 2 of our model). Column 4 reports similar estimates when we add interaction terms for types and illiquid assets rather than splitting the sample, although the estimates of the interaction terms are not significant.

Column 5 and 6 report qualitatively similar results to column 1 and 4 when we use retirement age as the outcome variables. However, the estimates are less precise and we do not split the sample by illiquid asset holding because – as explained above – the sub-sample of retirees who reports illiquid assets in 2016 is particularly small. Still, we find that naïve people retire 1.91 years earlier than time-consistent when they do not hold illiquid asset, but that illiquid assets make them postpone retirement by 1.37 years. Conversely, fully sophisticated people retire 1.73 years later than naïve when they do not hold illiquid asset, but the difference is smaller (+1.73-0.56) when we compare people with illiquid assets. While these estimates are in line with our hypothesis, they are not always precise.

Finally, taken together, our results suggest that partially sophisticated people under-estimate their present-bias to the extent that the cost of committing outweighs its perceived benefits. We then measure their sophistication level as the share of the self-control problem they are aware of ($\frac{predicted\ weight - ideal\ weight}{actual\ weight - ideal\ weight}$, see Augenblick and Rabin 2019). Their average sophistication level is 0.5, which suggests that being aware of half the size of the bias is not enough to induce different behaviour compared to being completely unaware of it.

VARIABLES	(1) Saving plan in 2016	(2) Private pension in 2012	(3) Labour force in 2020	(4) Retirement age
TI	-0.00	-0.03	-0.06**	-1.70**
	(0.03)	(0.09)	(0.03)	(0.69)
PS	0.04	0.15	0.01	0.35
	(0.03)	(0.10)	(0.03)	(0.82)
FS	0.09**	0.21**	0.08**	1.66**
	(0.04)	(0.09)	(0.03)	(0.81)
Observations	1,123	198	1,123	413
R-squared	0.19	0.31	0.45	0.39
Controls	Yes	Yes	Yes	Yes
Sample	Main	Main	Main	Retirees

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 3: Regression estimates for hypotheses 1 and 2.

Note: TI indicates being time-inconsistent, PS (FS) being partially (fully) sophisticated. Controls include gender, year of birth, weight, height, health status, monthly savings, patience, crystallized intelligence, state FE, and month of interview FE.

VARIABLES	(1) Labour force in 2020	(2) Labour force in 2020	(3) Labour force in 2020	(4) Labour force in 2020	(5) Retirement age	(6) Retirement age
TI	-0.06**	-0.01	-0.08**	-0.07**	-1.71**	-1.91**
	(0.03)	(0.06)	(0.04)	(0.04)	(0.69)	(0.76)
PS	0.00	0.01	0.00	0.01	0.36	0.34
	(0.03)	(0.05)	(0.03)	(0.03)	(0.82)	(0.88)
FS	0.07**	0.00	0.09**	0.08**	1.65**	1.73*
	(0.03)	(0.06)	(0.04)	(0.04)	(0.81)	(0.96)
TI x Saving plan				0.05		1.37
				(0.07)		(1.77)
PS x Saving plan				-0.01		0.17
				(0.06)		(1.93)
FS x Saving plan				-0.05		-0.56
				(0.07)		(1.73)
Saving plan in 2016	0.09***			0.07	0.35	-0.49
	(0.03)			(0.05)	(0.71)	(1.36)
Observations	1,123	298	825	1,123	413	413
R-squared	0.46	0.45	0.47	0.46	0.39	0.39
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Main	With saving plan in 2016	Without saving plan in 2016	Main	Retirees	Retirees

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 4: Regression estimates for hypothesis 3.

Note: TI indicates being time-inconsistent, PS (FS) being partially (fully) sophisticated. Controls include gender, year of birth, weight, height, health status, monthly savings, patience, crystallized intelligence, state FE, and month of interview FE.

5 Conclusion

This paper extends the analysis of retirement decisions with present-biased preferences by studying the role of illiquid assets as commitment devices, building on insights from Laibson (1997) and Diamond and Kőszegi (2003). A simple model shows that sophisticated individuals use illiquid assets not only to influence future consumption but also to prevent retiring earlier than planned. This ability to commit leads sophisticated agents to reach both their first-best consumption levels and retirement age.

Empirical evidence using population-representative German data supports the main model prediction: Naïve present-biased individuals retire on average 1.7 years earlier than time-consistent, while fully sophisticated individuals retire 1.7 years later than naïve, thereby fully overcoming their bias. We also find that fully sophisticated people are more likely to hold illiquid assets, and that illiquid assets are positively correlated with later retirement even after conditioning on types. Moreover, differences in retirement timing across types are smaller among people with illiquid assets, but remain large among those without. These results highlight the role of illiquid assets as commitment devices to influence retirement decisions.

Our results have important implications for policy. Retirement planning tools and policies that encourage the use of illiquid assets, such as pension plans with restricted early access or automated saving schemes, may help people achieve their long-term goals both in the saving and labour supply domain. Moreover, insights into the determinants of sophistication may inspire new policies to extend working lives, complementing traditional ones.

6 Appendix

6.1 Case 1 of the model

Self 1 and self 2 agree to work given saving s_1^w and no commitment, that is $U^w(s_1^w) = U(s_1^w) \geq U(s_1^r)$, meaning that the following holds (for any λ, see Section 2.2):

$$\ln(\lambda(s_1^w + w_2)) - \ln(\lambda s_1^w) + \beta[\ln((1-\lambda)(s_1^w + w_2)) - \ln((1-\lambda)s_1^w)] \geq e \tag{21}$$

Self 1 sets α such that

$$\begin{aligned} \lambda_1(s_1^w + w_2) &= \frac{1}{2}(s_1^w + w_2) = \alpha^w s_1^w + w_2 \\ \alpha^w &= \frac{\frac{1}{2}(s_1^w + w_2) - w_2}{s_1^w} < \lambda_1 = \frac{1}{2} \end{aligned}$$

and the liquidity constraint (LC) in (13) binds by construction. Conditional on working, self 2 is liquidity constrained and consumes $\alpha^w s_1^w + w_2 = \lambda_1(s_1^w + w_2)$ in period 2 and $(1-\lambda_1)(s_1^w + w_2)$ in period 3. Because the LC conditional on working (13) binds, the LC conditional on retiring (12) also binds:

$$\begin{aligned} \frac{1}{1+\beta}(s_1^w + w_2) &> \alpha^w s_1^w + w_2 \\ \frac{1}{1+\beta}(s_1^w + w_2) - w_2 &> \alpha^w s_1^w + w_2 - w_2 \\ \frac{1}{1+\beta}s_1^w - \frac{\beta}{1+\beta}w_2 &> \alpha^w s_1^w \\ \frac{1}{1+\beta}s_1^w &> \alpha^w s_1^w \end{aligned}$$

Thus, conditional on retiring, self 2 is liquidity constrained and consumes $\alpha^w s_1^w$ in period 2 and $(1-\alpha^w)s_1^w$ in period 3. Self 2 works if

$$\ln(\lambda_1(s_1^w + w_2)) - \ln(\alpha^w s_1^w) + \beta[\ln((1-\lambda_1)(s_1^w + w_2)) - \ln((1-\alpha^w)s_1^w)] \geq e \tag{22}$$

Because (i) (21) holds, (ii) $\alpha^w < \lambda_1 < \lambda_2$, and (iii) self 2 utility is concave in λ with λ_2 being optimum, then (22) is satisfied and self 2 works given s_1^w and α^w.

6.2 Case 2 of the model

Self 1 and self 2 agree to retire given self 1's first-best saving level s_1^r and no commitment, which occurs when $w_1/w_2 > ub$. We show here that, conditional on self 1 saving s_1^r and investing a share $\alpha^r = 1/2$ in the liquid asset, self 2 still prefers to retire when $w_1/w_2 > ub$.

Conditional on retiring, the LC of self 2 binds by construction. However, if self 2 work, the LC does not necessarily bind (this happens when $w_1/w_2 \leq (1+2\beta)/(1-\beta) \equiv \phi$, which can occur when $w_1/w_2 > ub$ as shown in Figure 2). We thus focus on the case in which the LC of self 2 does not bind if he works. In fact, if self 2 prefers to retire when working allows to break the LC, it follows that he also prefer to retire if working does not allow to break the LC.

If the LC does not bind, self 2 works if

$$\ln(\lambda_2(s_1^r + w_2)) + \beta \ln((1-\lambda_2)(s_1^r + w_2)) - \ln(a^r s_1^r) - \beta \ln((1-\alpha^r)s_1^r) \geq e$$

which, after substituting, boils down to

$$\frac{w_1}{w_2} \leq \frac{1}{\left(\exp\left(\frac{1-\beta\ln(\beta)}{1+\beta}\right) - \frac{2}{1+\beta}\right)\frac{(1+\beta)\beta}{2\beta+1}} \equiv \gamma. \tag{23}$$

But $\frac{w_1}{w_2} \leq \gamma$ contradicts the two conditions on the ratio w_1/w_2, namely $w_1/w_2 > ub$ and $w_1/w_2 \leq \phi$ (see Figure 2), meaning that self 2 would not change his decision and work if self 1 uses the commitment device.

6.3 Case 3 of the model

Self 1 and self 2 disagree on the retirement decision given self 1's first-best saving level s_1^w and no commitment, which happens when $lb < w_1/w_2 \leq ub$. Suppose that self 1 saves s_1^w and invest a share α^w in the liquid asset. Section 6.1 shows that the LC of self 2 binds both if he works and if he retires in period 2. Thus self 2 works if

$$\ln(\lambda_1(s_1^w + w_2)) - \ln(\alpha^w s_1^w) + \beta[\ln((1-\lambda_1)(s_1^w + w_2)) - \ln((1-\alpha^w)s_1^w)] \geq e \tag{24}$$

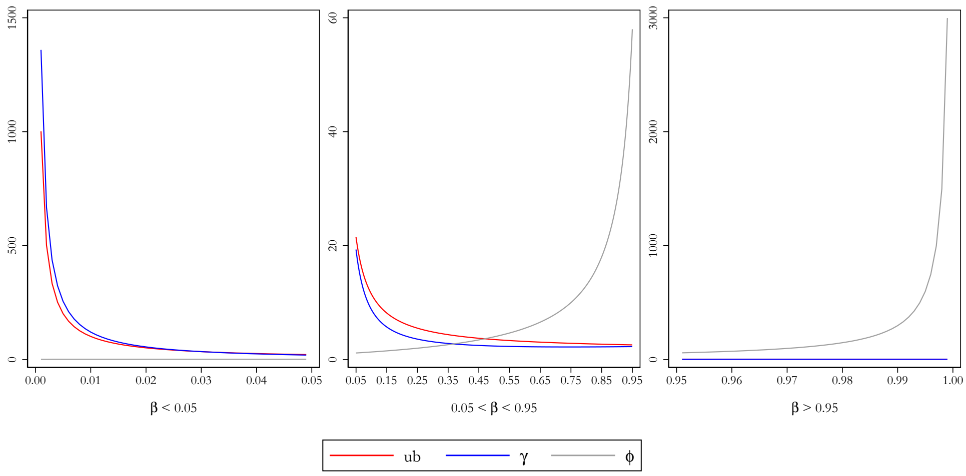

Figure 2: Bounds on w_1/w_2 for case 2 as a function of β, see Section 6.2.

First, note that $\alpha^w s_1^w$ needs not to be positive. If $\alpha^w s_1^w \leq 0$, self 2 works. If $\alpha^w s_1^w > 0$, after substituting, (24) boils down to

$$\frac{w_1 + w_2}{\beta w_1 - w_2 - \beta w_2} \geq \exp(e - \ln(\beta)).$$

Define the ratio $r \equiv w_1/w_2$ and substitute $w_1 = rw_2$

$$\frac{r+1}{\beta r - 1 - \beta} \geq \exp(e - \ln(\beta)). \tag{25}$$

Because $1 - \beta \exp(e - \ln(\beta)) = 1 - \exp(e) < 0$, (25) gives

$$r \equiv \frac{w_1}{w_2} \leq \frac{-(1+\beta)\exp(e - \ln(\beta)) - 1}{1 - \beta \exp(e - \ln(\beta))} \equiv \delta. \tag{26}$$

Figure 3 shows that $ub < \delta$, meaning that condition (26) is satisfied whenever $lb < w_1/w_2 \leq ub$. Therefore, self 1 choses s_1^w and α^w, and self 2 works.

6.4 A simple extension with uncertainty

We now assume that the agent receives income $W > 0$ in period 1, while income in period 2 is an exogenous random variables U which takes value $\overline{U}$ with probability p and $\underline{U}$ with probability $1 - p$, with $\overline{U} > \underline{U} \in \mathbb{R}$ and $p \in [0, 1]$. We abstract from labour supply choices for the moment. This simple modelling of uncertainty encompasses, for example, income risk, unemployment

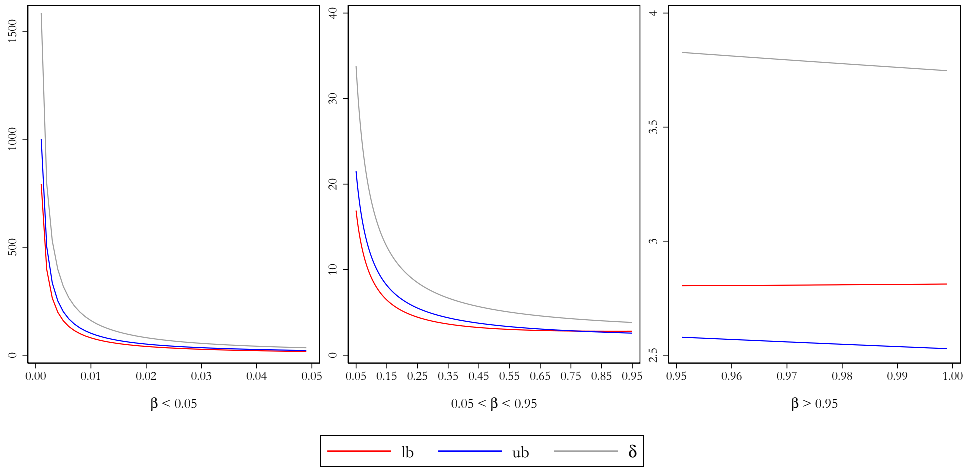

Figure 3: Bounds on w_1/w_2 for case 3 as a function of β, see Section 6.3.

risk, and uncertain future expenses. Present-biased self 1 then solves

$$\max_{c_1} u(c_1) + p\beta\{u(\lambda(W - c_1 + \overline{U})) + u((1-\lambda)(W - c_1 + \overline{U}))\} +$$
$$+(1-p)\beta\{u(\lambda(W - c_1 + \underline{U})) + u((1-\lambda)(W - c_1 + \underline{U}))\}$$

where $\lambda = \lambda_2$ if self 1 is sophisticated and realizes that self 2 ultimately decides on the consumption allocation, while $\lambda = \lambda_1$ corresponds to the first-best allocation. It's then easy to see that the change in life-time utility associated with moving from the second-best to the first-best consumption allocation for self 1 is still given by equation (2), as for the case without uncertainty. Therefore, as in the main text, sophisticated self 1 would stick to the same consumption/saving level while trying to improve on the future consumption allocation.

Assume now that an illiquid asset is available, with the same characteristics discussed in the main text. If $p = 1$ ($U = \overline{U}$), self 1 chooses an optimal share of liquidity $\overline{\alpha} < 1$. If $p = 0$ ($U = \underline{U}$), self 1 chooses a different share $\underline{\alpha} < 1$ (with $\overline{\alpha} < \underline{\alpha}$, i.e. more liquidity in the 'bad' state). Also when $p \in (0, 1)$, not using the commitment ($\alpha = 1$) is a dominated strategy. For example, $\underline{\alpha}$ is preferred to $\alpha = 1$. In particular, with $\underline{\alpha}$ the commitment will not bind if $\overline{U}$ realizes (such that self 2 consumes a share λ_2), but it will lead to the first-best allocation if $\underline{U}$ realizes (corresponding to λ_1). Therefore, the utility is the same in the 'good' state but improved in the 'bad' one due to the commitment, leading to higher expected utility. The same argument applies if U takes more than two values: targeting the commitment to the worst-case scenario is always better than no commitment, because the allocation in the other states is not altered. However, because the commitment cannot be targeted to every realization of the state space, the first-best allocation is not always attainable. Therefore, the marginal utility of commitment is still strictly positive under uncertainty, although smaller compared to the model without uncertainty. If the commitment comes at a positive direct cost, the agent might choose to commit without uncertainty but to not commit with uncertainty.

Also for the case of endogenous retirement, targeting the illiquid asset to the worst-case scenario is always better than no commitment. Suppose, for example, that self 1 and self 2 agree to work if $\overline{U}$ realizes and agree to retire if $\underline{U}$ realizes (the opposite cannot happen). If self 1 targets the commitment to the worst-case and sets $\alpha = 1/2$ (see Section 3), the consumption allocation in the other state is not altered because the commitment will not bind (as above) and also the work decision is not altered (same logic of Section 6.1, if they agree to work given $\alpha = 1$ they also agree with less liquidity). The same logic applies for the other possible cases, e.g. if self 1 and self 2 agree to work if $\overline{U}$ realizes but disagree if $\underline{U}$ realizes (the opposite cannot happen), or if self 1 and self 2 disagree if $\overline{U}$ realizes but agree to retire if $\underline{U}$ realizes (the opposite cannot happen), etc.

6.5 Additional information on the variable definitions

Table 5 provides information on the variable definitions and reports the year in which they were assessed. When the year of assessment is particularly important for our empirical application, we include it also in the variable name.

Variable	Definition
Weight in 2017	Respondent's body weight in kg in 2017.
Weight in 2018	Respondent's body weight in kg in 2018.
Ideal weight	Answer to "What weight (in kg) do you view as ideal for yourself at the time of the next interview in approximately one year?". The variable is assessed in 2017.
Predicted weight	Answer to "What weight (in kg) do you expect to have at the time of the next interview in approximately one year?". The variable is assessed in 2017.
Saving plan in 2016	Dummy equals 1 if the respondent held any of the following: personal pension schemes with state subsidies (Riester- or Rürup-Rente), other personal pension schemes, building savings contracts, cash-value life insurance policies, or capital formation savings plans. 0 otherwise. The variable is assessed in 2016.
Private pension in 2012	Dummy equals 1 if the respondent held a personal pension schemes with state subsidies (Riester- or Rürup-Rente). 0 otherwise. The variable is assessed in 2012.
Retirement age	Age of retirement computed as last year of employment minus year of birth. If the last year of employment is not available, we construct it using information about labour force participation in waves from 2012 to 2020.
Labour force in 2020	Dummy equals 1 if the respondent is in the labour force, 0 otherwise. The variable is assessed in 2020.
Female	Dummy equals 1 if the respondent is female, 0 otherwise.
Year of birth	Respondents' year of birth.
Monthly savings	Answer to "Do you usually have money left over at the end of the month that you can put aside for larger purchases, emergencies, or to build savings? If so, how much?" The variable is assessed in 2016 and expressed in EUR.
Height	Respondent's body height in cm in 2017.
Health status	Answer to "How would you describe your current state of health?" on a 5-point scale. The variable is assessed in 2017.
Patience	Answer to "Would you describe yourself as an impatient or a patient person in general?" on an 11-point Likert scale, standardized to mean 0 and standard deviation 1. The variable is assessed in 2013.
Crystallized intelligence	Respondents are asked to name as many animals as possible in 90 seconds. The test score is the number of uniquely named animals within the time span, see Lang et al. (2007). The test score is standardized to mean 0 and standard deviation 1. The variable is assessed in 2014.
State of residence	Respondent's federal state of residence. Enters regressions as control variable through fixed effects. The variable is assessed in 2017.
Month of interview	Month respondent is interviewed. Enters regressions as control variable through fixed effects. The variable is assessed in 2017.

Table 5: Variable definitions.

6.6 Comparison of the classification results with the literature

In Table 6, we compare the result of our classification to studies that classify people at the *individual level* as time-consistent, naïve, or sophisticated. Several studies estimate average values for parameters related to present-bias and sophistication or they only classify people as time-consistent vs. time-inconsistent. These studies are not considered in the comparison.

Study	Sample size	Share of time-consistent people	Share of present-biased people with some sophistication
Wong (2008) (sample 1)	158	12.7%	73.4%
Wong (2008) (sample 2)	287	8.4%	70.7%
Cerrone and Lades (2017)	73	32.9%	42.5%
Mandel et al. (2017) (study 1A)	205	19.0%	70.0%
Mandel et al. (2017) (study 1B)	290	39.0%	36.0%
Mandel et al. (2017) (study 2)	218	12.0%	30.5%
Mandel et al. (2017) (study 3)	164	38.3%	44.7%
Mahajan et al. (2023)	566	32.0%	18.9%
Weighted average of listed studies by sample size	1,961	25.0%	56.1%
This paper	1,123	25.0%	56.0%

Table 6: Comparison of classification results.

Note: The samples in Wong (2008); Cerrone and Lades (2017); Mandel et al. (2017) only comprise university students, while Mahajan et al. (2023) studies households in rural India.

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